Conditional Generation with Classically Trainable IQP Circuits

Oriol Balló-Gimbernat[1,2,3] and Emiliano Tolotti[4]

[1]Eurecat, Centre Tecnològic de Catalunya, Barcelona, Spain
[2]Centre de Visió per Computador (CVC), Barcelona, Spain
[3]Universitat Autònoma de Barcelona (UAB), Barcelona, Spain
[4]Department of Information Engineering and Computer Science, University of Trento, Via Sommarive 9, 38123 Trento, Italy

Conditional generative models represent families of distributions indexed by an input variable. Conditioning instantaneous quantum polynomial-time (IQP) Born machines is constrained by their diagonal commuting gates and by classical training's reliance on a tractable reference state. We introduce an input-state-conditioned IQP Born machine in which the condition changes only the prepared input, while a compiled IQP unitary and its phase vector are shared across the family. Classical training remains efficient provided the state entering the diagonal unitary supports efficient basis sampling and amplitude-ratio evaluation. For product inputs, the amplitude-ratio estimator has second moment bounded by one, allowing classical estimation of the Pauli characters of a Hamming-kernel maximum mean discrepancy loss. This does not imply efficient output sampling or guaranteed optimization.

A 1,020-fit Gaussian campaign spans 8–19 qubits, three widths, and unseen conditions. Conditioning lowers profile error in every size–width cell, but circuits trained at individual conditions are usually more accurate. An 80-fit Curie–Weiss campaign tests a temperature-driven change from a central to a bimodal magnetization profile. A fixed 13-qubit circuit with five marginalized ancillas and a spin-flip-symmetric input follows this response and reduces held-out profile and raw error relative to no conditioning. In this campaign, the shared circuit matches or improves on condition-specific references, although the benchmark does not identify which architectural feature causes the improvement. Together, the campaigns show that input-state conditioning preserves the classical estimators and can express nontrivial conditional responses, while the cost of sharing depends on the target and input-state family.

Oriol Balló-Gimbernat: oriol.ballo@eurecat.org
Emiliano Tolotti: emiliano.tolotti@unitn.it

1 Introduction

Conditional generative models learn $q_\theta(x \mid c)$, a family of output laws selected by a condition c. Quantum circuit Born machines represent discrete laws through measurement statistics and have been trained with likelihood-free objectives such as maximum mean discrepancy (MMD) [1–3]. Earlier conditional QCBMs inject an external variable through repeated rotations or dedicated condition qubits [4, 5]. They show that QCBMs can be conditional. Our question is narrower. Can conditioning be added to an IQP Born machine while preserving its classical training framework?

Classically trained IQP Born machines exploit a separation between observable estimation and output sampling. Selected Pauli observables, and therefore character-kernel MMD losses, can be estimated classically for a tractable reference state even though other IQP families are associated with conditional sampling-hardness results [6–9]. This paper uses *classically trainable* only for the declared loss and gradient estimators. It does not claim classical access to the full output law.

We encode c in a classically tractable input state and then apply one fixed IQP unitary. A product input retains the factorized reference distribution used by the Pauli estimator; a non-product input remains usable whenever it supports exact basis sampling and amplitude ratios. Different input angles make the measured law

depend on c. The ordered diagonal generators, learned phases, and compiled IQP gates do not change with the condition. Figure 1 shows this separation. Input-dependent rotations and their finite Fourier response are established in quantum circuit learning [10–12]; our contribution is the formulation that places this encoding inside the classically observable-trainable IQP setting.

We validate the formulation in two cases. Gaussian Hamming-weight targets isolate translation and interpolation. Curie–Weiss targets change shape with inverse temperature and therefore test more than a moving peak. In both cases, a no-condition model measures whether the input encoding provides usable control, while independently trained circuits expose the accuracy cost of sharing one circuit. For Curie–Weiss, a visible-only model with nearly the same parameter count tests whether the ancilla result can be reproduced by higher-local visible phases.

The evidence is deliberately finite and descriptive. It concerns within-range interpolation, exchangeable targets, and the tested estimator budgets. We make no claim of universal conditional approximation, efficient classical output sampling, quantum advantage, thermal-state preparation, or a generally favorable optimization landscape. Recent IQP trainability analyses give further reason to keep the last boundary explicit [13].

2 Conditional IQP model

Let $g_j \in \{0,1\}^n$ specify a diagonal Pauli string $Z_{g_j} = \bigotimes_i Z_i^{g_{j,i}}$. For a fixed ordered generator bank $G = (g_1, \dots, g_m)$ and phases $\theta \in \mathbb{R}^m$, define

$$D_Z(\theta) = \prod_{j=1}^{m} \exp(i\theta_j Z_{g_j}). \tag{1}$$

The input-state-conditioned output law is

$$q_\theta(x \mid c) = \left| \langle x | H^{\otimes n} D_Z(\theta) H^{\otimes n} \times \bigotimes_{i=1}^{n} R_Y(\alpha_i(c)) |0\rangle \right|^2. \tag{2}$$

The preparation rotations precede the IQP unitary, so the complete pipeline is not itself a strict IQP circuit. The deployment contract is instead precise: $D_Z(\theta)$, G, and θ are identical for every c. Only the input state changes.

The Gaussian experiment uses a broadcast map

$$\alpha_i(c) = \pi(0.35 - 0.70c), \qquad i = 1, \dots, n, \tag{3}$$

and all one- and two-body diagonal generators, giving $m = n(n+1)/2$ learned phases. For an affine angle map, each output probability has finite Fourier support contained in signed sums of the encoding frequencies [11, 12]. This describes the available conditional response but does not imply universality.

The Curie–Weiss model has eight visible qubits and $r = 5$ ancillas. Its visible law is

$$q_\theta(x \mid \beta) = \sum_{z \in \{0,1\}^r} |\langle x, z | H D_Z(\theta) H | \phi(\beta) \rangle|^2, \tag{4}$$

where tensor powers of H are implicit and $|\phi(\beta)\rangle$ is the spin-flip two-branch state $|p(\beta)\rangle^{\otimes n} + X^{\otimes n}|p(\beta)\rangle^{\otimes n}$ on all 13 qubits, defined below. Tracing out the ancillas produces a mixed visible state with rank at most 2^r and raises the complete two-local generator count from 36 to 91. These are separate capacity changes. Pure visible states can already produce multimodal classical measurement laws and nonzero Shannon entropy, so ancilla marginalization is not necessary for either property.

Two reference families clarify the sharing tradeoff. A *no-condition* model receives the same neutral input for every target. A *single-condition reference* trains an independent circuit for one condition. At unseen midpoints we also evaluate a target-aware choice between adjacent references and a target-blind equal or barycentric mixture. The references use more stored phases and optimization work than one shared model; they measure attainable fidelity rather than parameter-matched efficiency.

3 Classical MMD training

For the Hamming radial-basis kernel $k_\sigma(x, y) = \exp[-d_H(x, y)/(2\sigma^2)]$, draw a mask $a \in \{0,1\}^n$ with independent coordinates

$$a_i \sim \text{Bernoulli}(q_\sigma), \qquad q_\sigma = \frac{1 - e^{-1/(2\sigma^2)}}{2}. \tag{5}$$

The kernel has the character representation

$$k_\sigma(x, y) = \mathbb{E}_a[(-1)^{a \cdot x}(-1)^{a \cdot y}], \tag{6}$$

so MMD is an average of discrepancies between Pauli-Z characters [14]. Averaging the unbiased

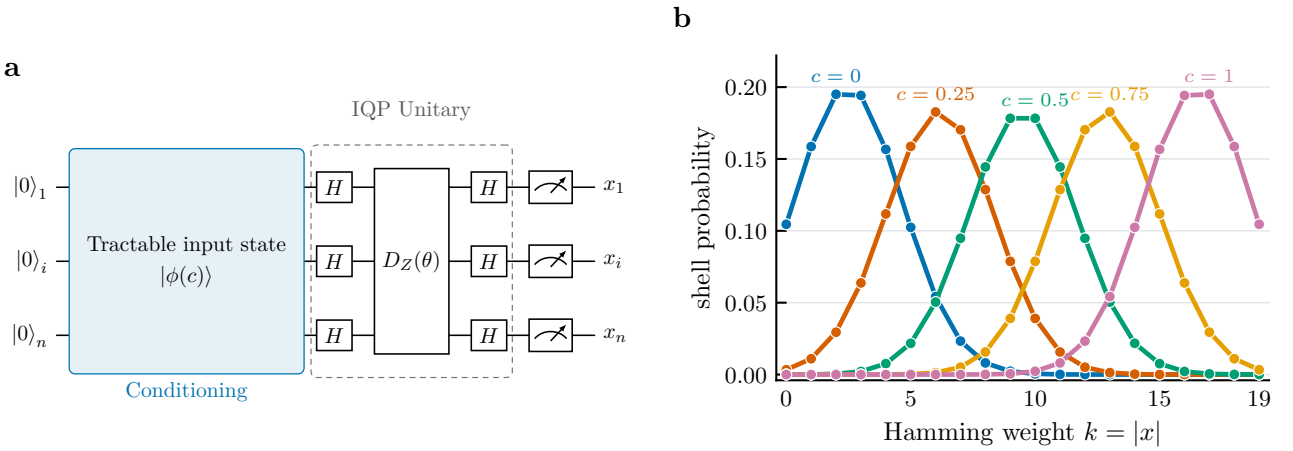


Figure 1: Input-state conditioning and the Gaussian target family. (a) Conditioning prepares a classically tractable input state $|\phi(c)\rangle$ (a product state in the Gaussian experiment and a spin-flip two-branch state in the Curie–Weiss experiment); the IQP unitary and phase vector are shared across conditions. (b) Five representative 19-qubit target profiles; probability is uniform within each Hamming shell.

per-condition U-statistics gives the conditional training loss. Automatic differentiation of the sampled finite expression supplies phase gradients.

Product structure is sufficient for classical estimation, but it is not necessary. Write $\phi_c(z) = \langle z|H^{\otimes n}|\phi(c)\rangle$ for the state entering the diagonal unitary. Uniformly over c, the estimator requires a classical algorithm that can sample z from $|\phi_c(z)|^2$ and evaluate the complex ratio $R_c(z,a) = \phi_c(z \oplus a)/\phi_c(z)$ to controlled precision in polynomial time. A sampling oracle alone is insufficient because each Monte Carlo draw must be assigned this ratio. Product states provide both operations by factorizing over qubits; sample-only implicit models and generic circuit states do not provide them without additional structure. Zero-amplitude strings are never sampled, so the ratio need only be defined on the support of $|\phi_c|^2$. Van den Nest's computationally tractable-state condition, which combines sampling with amplitude queries, is sufficient for these operations [6]. Sampling z from $|\phi_c(z)|^2$ then gives

$$\mathbb{E}_z\left[|R_c(z,a)|^2\right] = \sum_{z:\phi_c(z)\neq 0} |\phi_c(z \oplus a)|^2 \leq 1. \quad (7)$$

Therefore the relevant estimator has a bounded second moment and variance at most one. This is not a zero-variance result. It also does not guarantee that gradient-based optimization succeeds or scales favorably.

The Gaussian campaign samples 64 masks during training and 256 during validation. The Curie–Weiss campaign integrates all 256 visible masks, padded by identity on the ancillas. Both use 8,192 model samples in training, 16,384 in validation, bandwidths $(1, 2, 4)$, and checkpoints selected by validation conditional MMD. Exact probabilities are enumerated only after checkpoint selection. The supplement gives the estimator derivation and a finite small-system consistency and resource audit.

4 Gaussian-mixture experiments

The Gaussian target assigns a truncated discrete Gaussian to Hamming weight $w \in \{0, \ldots, n\}$ and spreads each shell mass uniformly over its $\binom{n}{w}$ strings. Its requested mean is $n(0.15 + 0.70c)$ and its nominal width is $s\sqrt{n}$, with $s \in \{0.35, 0.50, 0.75\}$. Profile TVD compares the target and generated Hamming-weight laws. Raw bitstring TVD records within-shell errors that profile TVD omits.

The factorial study crosses $n \in \{8, 12, 16, 19\}$, three widths, and ten paired seeds. Shared conditioned and no-condition circuits train at five values of c and are evaluated at eight unseen odd-sixteenth coordinates. Five single-condition circuits per seed supply the references. Every fit uses 2,000 Adam updates with cosine decay from 0.01 to 0.001. The factorial and condition-count studies contain 1,020 immutable fits.

Conditioning improves training-condition profile TVD over no conditioning in all 12 size–width cells. At $n = 19, s = 0.50$, the largest sys-

tem at the middle width, the paired improvement is 0.2668 [0.2587, 0.2737]. The independent-circuit fidelity gap is 0.0878 [0.0860, 0.0897], with positive values indicating that the separate circuits are more accurate. At unseen conditions, the target-aware adjacent-reference comparison is 0.0016 [-0.0012, 0.0043] and the coordinate-weighted mixture comparison is -0.0029 [-0.0053, -0.0006]. Across the complete surface, the adjacent reference is better in ten cells and mixed in two; the mixture is better in nine, mixed in one, and worse in two. Product-state conditioning therefore provides reliable control relative to no condition, but sharing one circuit has a measurable fidelity cost.

The condition-count study separates denser coverage from additional data at $n = 19, s = 0.50$. With a fixed number of samples per condition, mean held-out profile TVD falls from 0.382 at $K = 2$ to 0.167 at $K = 5$ and 0.160 at $K = 9$. Holding total data fixed gives 0.382, 0.165, and 0.162 at the same grid sizes. The near agreement shows that covering the condition interval matters more than adding samples at conditions already present; improvement is small beyond $K = 5$ on this task.

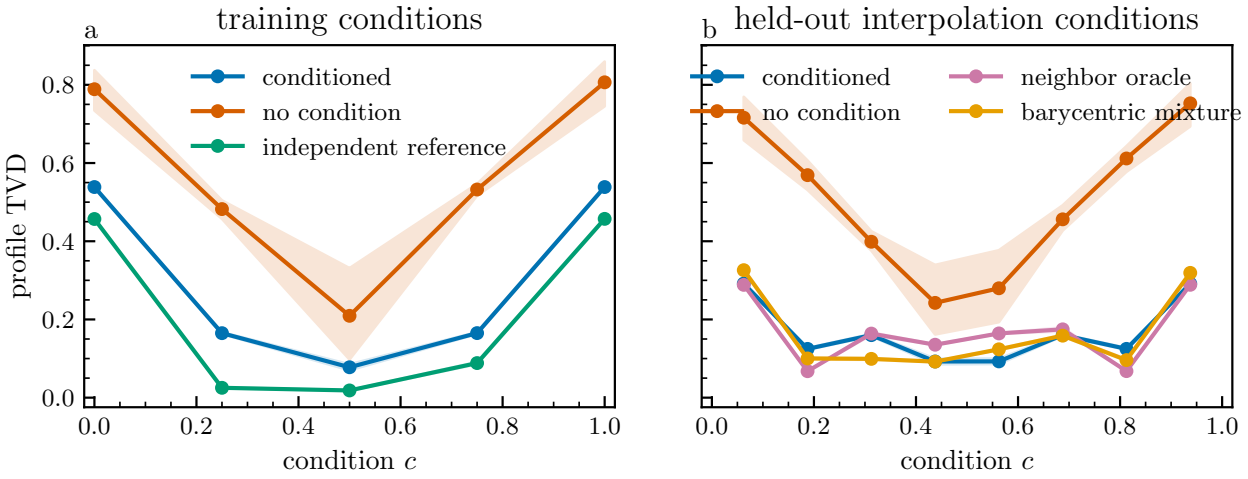


Figure 2: Gaussian conditional control at $n = 19, s = 0.50$, the largest system at the middle width and the cell used for the condition-count study. Lines are ten-seed means and bands are deterministic 95% paired-bootstrap intervals. Training results compare the conditioned and no-condition models. Held-out results include adjacent single-condition references and their coordinate-weighted mixture. The complete 12-cell factorial appears in the supplement.

5 Curie–Weiss experiments

For spins $s_i \in \{-1, +1\}$, $M = \sum_i s_i$, and eight visible sites, the target law is

$$p_\beta(s) = Z_\beta^{-1} \exp\left(\frac{\beta M^2}{16}\right). \tag{8}$$

It is invariant under spin flip and qubit permutations. Increasing β changes the finite-size magnetization profile from central to bimodal. We use profile and raw bitstring TVD as co-primary metrics. The unseen $\beta = 1$ point is a finite-size crossover probe, not an estimate of critical scaling.

Six training temperatures, $\{0.5, 0.7, 0.9, 1.1, 1.3, 1.5\}$, alternate with five held-out midpoints. The production encoder is a broadcast piecewise linear map with anchors

$$\frac{\alpha(\beta)}{\pi} = (-0.14, -0.07, 0, 0.08, 0.14) \tag{9}$$

at $\beta = (0.5, 0.9, 1.0, 1.2, 1.5)$ and linear interpolation between them. We selected these knots using development screens at training temperatures and froze them before the 80 production fits. No production outcome was used to retune the map. The input state is the spin-flip two-branch superposition

$$|\phi(\beta)\rangle \propto |p(\beta)\rangle^{\otimes n} + X^{\otimes n}|p(\beta)\rangle^{\otimes n}, \tag{10}$$

with $|p(\beta)\rangle = R_Y[\alpha(\beta)]|0\rangle$ on all 13 qubits. Its post-Hadamard amplitude depends only on the Hamming weight, so the exact reference sampling and amplitude ratios used in training remain classically tractable, while the two branches encode

the $m \leftrightarrow -m$ symmetry of the zero-field target. This spin-flip-symmetric input follows the global-bitflip ansatz of Ref. [8] (Sec. 6 therein), here made condition-dependent through $\alpha(\beta)$.

Ten paired seeds produce ten conditioned models, ten no-condition controls, and sixty single-temperature references. All use five ancillas and one 13-qubit complete two-local circuit with 91 phases and 2,000 updates. The exact eight-qubit visible marginal is evaluated after checkpoint selection. Conditions are averaged within seed before a 10,000-resample paired bootstrap.

The conditioned model improves profile TVD over no conditioning by +0.134 [+0.129, +0.140] at training temperatures and +0.121 [+0.116, +0.127] at held-out temperatures, and raw TVD by +0.140 [+0.135, +0.145] and +0.128 [+0.123, +0.134]. The effect is positive in the critical region $\beta \in \{0.9, 1.0, 1.1\}$ (+0.038 [+0.030, +0.047] profile, +0.056 [+0.048, +0.065] raw) and strongest in the cold region $\beta \in \{1.3, 1.4, 1.5\}$. The shared circuit matches the adjacent mixture (-0.011 [-0.017, -0.005] profile, +0.008 [+0.002, +0.013] raw) and beats the target-aware adjacent oracle (+0.015 [+0.011, +0.019] profile, +0.036 [+0.031, +0.041] raw). For this finite campaign, the two-branch input therefore yields a condition-dependent temperature response without a resolved fidelity loss relative to the condition-specific references.

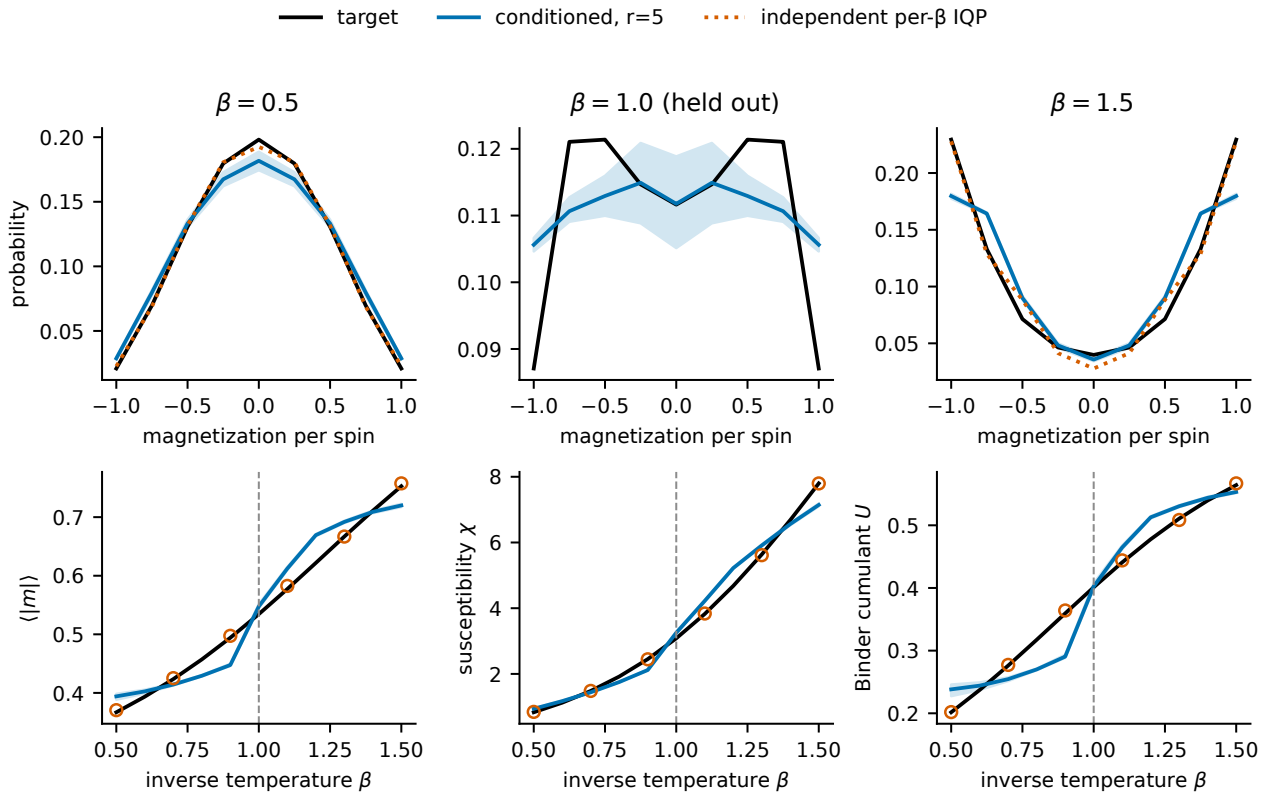


Figure 3: Curie–Weiss temperature response of the fixed 13-qubit model. The top row shows magnetization profiles at the hot endpoint, unseen $\beta = 1$ midpoint, and cold endpoint. The bottom row reports response observables. Bands summarize ten matched seeds. Orange circles show the independently trained per-β IQP circuits at the six training temperatures; this reference is not defined at the five held-out temperatures.

6 Discussion

The experiments show what input-state conditioning buys and what it does not. One learned phase vector and one compiled IQP unitary respond to an external variable without condition-dependent diagonal gates. Across both benchmarks, this response improves over withholding the condition. Separate circuits remain more accurate in the Gaussian comparisons, while no such gap is resolved in the Curie–Weiss v3 campaign. The shared model therefore exchanges fidelity for fixed compiled structure, constant stored parameters, and constant optimization count as the condition grid grows.

Several limitations are structural. Both core targets use a one-dimensional condition and are

exchangeable. Profile TVD ignores within-shell variation, which remains visible in raw TVD. The Curie–Weiss evidence covers eight visible spins, one target family, and within-range temperatures. Ancillas change both the visible-state rank and the interaction structure, and no matched-capacity control was run for the v3 spin-flip input. A historical v2 control and two post-core symmetry screens are reported in the supplement as diagnostics, not as causal or confirmatory evidence for the v3 result.

The classical-training claim is equally bounded. Equation (7) controls the estimator's second moment, not the optimization landscape. Exact post-selection evaluation is exponential and is used only to audit these finite systems. We have not shown efficient classical sampling, quantum advantage, extrapolation, universal conditional expressivity, or thermal-state preparation.

Within those limits, the formulation is a useful base for further work. It preserves the observable interface of classically trained IQP Born machines while adding a condition through a simple input-state preparation. The Gaussian campaign maps the sharing tradeoff over size, width, and condition density. The Curie–Weiss campaign shows that the same mechanism can follow a shape-changing family. Larger visible systems, nonexchangeable targets, and encodings chosen independently of target outcomes are the next tests.

Data and code availability

The code, frozen campaign summaries, and manuscript sources are assembled in a local coauthor-review package. Public repository and archival identifiers will be added only after author approval. No upload or submission action is included.

References

[1] Jin-Guo Liu and Lei Wang. Differentiable learning of quantum circuit born machine. *Physical Review A*, 98(6):062324, 2018. DOI: 10.1103/PhysRevA.98.062324.

[2] Marcello Benedetti, Delfina Garcia-Pintos, Oscar Perdomo, Vicente Leyton-Ortega, Yunseong Nam, and Alejandro Perdomo-Ortiz. A generative modeling approach for benchmarking and training shallow quantum circuits. *npj Quantum Information*, 5:45, 2019. DOI: 10.1038/s41534-019-0157-8.

[3] Xun Gao, Zhengyu Zhang, and Luming Duan. A quantum machine learning algorithm based on generative models. *Science Advances*, 4(12): eaat9004, 2018. DOI: 10.1126/sciadv.aat9004.

[4] Oriel Kiss, Michele Grossi, Enrique Kajomovitz, and Sofia Vallecorsa. Conditional born machine for monte carlo event generation. *Physical Review A*, 106(2):022612, 2022. DOI: 10.1103/PhysRevA.106.022612.

[5] Qing-Wei Zeng, Hong-Ying Ge, Chen Gong, and Nan-Run Zhou. Conditional quantum circuit born machine based on a hybrid quantum–classical framework. *Physica A: Statistical Mechanics and its Applications*, 618:128693, 2023. DOI: 10.1016/j.physa.2023.128693.

[6] Maarten Van den Nest. Simulating quantum computers with probabilistic methods. *Quantum Information and Computation*, 11(9–10): 784–812, 2011.

[7] Erik Armengol and Joseph Bowles. IQPopt: Fast optimization of instantaneous quantum polynomial circuits in JAX, 2025.

[8] Erik Recio-Armengol, Shahnawaz Ahmed, and Joseph Bowles. Train on classical, deploy on quantum: Scaling generative quantum machine learning to a thousand qubits, 2025.

[9] Michael J. Bremner, Ashley Montanaro, and Dan J. Shepherd. Achieving quantum supremacy with sparse and noisy commuting quantum computations. *Quantum*, 1:8, 2017. DOI: 10.22331/q-2017-04-25-8.

[10] Kosuke Mitarai, Makoto Negoro, Masahiro Kitagawa, and Keisuke Fujii. Quantum circuit learning. *Physical Review A*, 98(3):032309, 2018. DOI: 10.1103/PhysRevA.98.032309.

[11] Maria Schuld, Ryan Sweke, and Johannes Jakob Meyer. The effect of data encoding on the expressive power of variational quantum-machine-learning models. *Physical Review A*, 103(3):032430, 2021. DOI: 10.1103/PhysRevA.103.032430.

[12] Berta Casas and Alba Cervera-Lierta. Multidimensional fourier series with quantum circuits. *Physical Review A*, 107(6):062612, 2023. DOI: 10.1103/PhysRevA.107.062612.

[13] Kevin Shen, Susanne Pielawa, Vedran Dunjko, and Hao Wang. Characterizing trainability of instantaneous quantum polynomial circuit born machines, 2026.

[14] Arthur Gretton, Karsten M. Borgwardt, Malte J. Rasch, Bernhard Schölkopf, and Alexander Smola. A kernel two-sample test. *Journal of Machine Learning Research*, 13:723–773, 2012. URL https://jmlr.org/papers/v13/gretton12a.html.

A Analytical details

A.1 Finite conditional response

Let the product-state angle map be affine, $\alpha(c) = b + Wc$, where row i of W is w_i^T. Expanding each local sine and cosine into exponentials shows that every amplitude has frequencies in signed sums of the rows of W. Squaring an amplitude gives

$$\Omega_{\text{full}}(W) \subseteq \left\{ \sum_{i=1}^{n} s_i w_i : s_i \in \{-1, 0, 1\} \right\}. \tag{11}$$

Coincident frequencies and vanishing coefficients can make the inclusion strict. The argument establishes a finite response space at fixed encoding width; it does not provide an efficient coefficient-evaluation procedure or a universal-approximation result.

For an observable Z_a, the active-generator expansion retains only generators with $a \cdot g_j = 1 \pmod 2$. Grouping active subsets by their XOR mask h and using $\langle Y_i \rangle = 0$ for a real R_Y product input removes terms with $h \wedge a \neq 0$. Observable responses can therefore have strictly smaller frequency support than full probabilities. This is a direct specialization of established circuit-Fourier and IQP active-generator descriptions [8, 11, 12].

A.2 Product and block-product estimator

Write the tractable reference amplitude as $\phi_c(z) = \langle z | H^{\otimes n} | \phi(c) \rangle$. The classical observable estimator samples z from $|\phi_c(z)|^2$ and evaluates an amplitude ratio $R_c(z, a) = \phi_c(z \oplus a)/\phi_c(z)$ together with the phase factor generated by the active diagonal terms. The ratio obeys

$$\mathbb{E}_z |R_c(z, a)|^2 = \sum_{z : |\phi_c(z)| > 0} |\phi_c(z)|^2 \frac{|\phi_c(z \oplus a)|^2}{|\phi_c(z)|^2} \tag{12}$$

$$\leq \sum_z |\phi_c(z \oplus a)|^2 = 1. \tag{13}$$

The equality used in the main text holds when the shifted support is fully covered; the inequality is the general statement when some reference amplitudes vanish. In either case the variance is at most one. The bound is on the estimator variance, not on optimizer convergence or raw gradient magnitude.

The same argument applies to a block-product input $|\phi(c)\rangle = \bigotimes_B |\phi_B(c)\rangle$. The ratio factorizes over blocks, and normalization gives

$$\mathbb{E} |R_c(z, a)|^2 \leq \prod_B \sum_{z_B} |\phi_B(z_B \oplus a_B)|^2 = 1. \tag{14}$$

This observation supports the archived block-encoding study in Sec. E. The Gaussian production experiments use single-qubit product states, while the Curie–Weiss production experiments use the spin-flip two-branch state, whose radial amplitude also factorizes over Hamming shells and is classically tractable.

A.3 Hamming-kernel character loss

For $a_i \sim \text{Bernoulli}(q_\sigma)$ with q_σ defined in Eq. (5),

$$\mathbb{E}_a[(-1)^{a \cdot (x \oplus y)}] = \prod_i [(1 - q_\sigma) + q_\sigma (-1)^{x_i \oplus y_i}] \tag{15}$$

$$= \exp[-d_H(x, y)/(2\sigma^2)]. \tag{16}$$

For condition set C, the training objective is the balanced average

$$\mathcal{L}(\theta) = \frac{1}{|C|} \sum_{c \in C} \text{MMD}^2(q_\theta(\cdot \mid c), p(\cdot \mid c)). \tag{17}$$

Table 1: Gaussian factorial profile-TVD effects with paired 95% bootstrap intervals. Status letters are S for supported, M for mixed, and U for unsupported, in the order no-condition, adjacent-reference, mixture.

n	s	no condition − conditioned	adjacent reference − conditioned	mixture − conditioned	status
8	0.35	+0.2402 [+0.2278, +0.2536]	-0.0292 [-0.0468, -0.0108]	-0.0248 [-0.0418, -0.0064]	S/U/U
8	0.50	+0.2170 [+0.2015, +0.2418]	-0.0280 [-0.0336, -0.0225]	-0.0363 [-0.0432, -0.0292]	S/U/U
8	0.75	+0.1460 [+0.1293, +0.1714]	-0.0276 [-0.0339, -0.0214]	-0.0525 [-0.0584, -0.0461]	S/U/U
12	0.35	+0.2419 [+0.2352, +0.2505]	-0.0221 [-0.0255, -0.0190]	-0.0016 [-0.0052, +0.0019]	S/U/M
12	0.50	+0.2291 [+0.2213, +0.2391]	-0.0180 [-0.0213, -0.0146]	-0.0251 [-0.0283, -0.0216]	S/U/U
12	0.75	+0.1608 [+0.1515, +0.1722]	-0.0129 [-0.0151, -0.0107]	-0.0548 [-0.0567, -0.0526]	S/U/U
16	0.35	+0.2472 [+0.2421, +0.2519]	-0.0229 [-0.0279, -0.0174]	+0.0117 [+0.0067, +0.0172]	S/U/S
16	0.50	+0.2440 [+0.2354, +0.2537]	-0.0040 [-0.0071, +0.0001]	-0.0099 [-0.0133, -0.0057]	S/M/U
16	0.75	+0.1667 [+0.1570, +0.1770]	-0.0192 [-0.0205, -0.0178]	-0.0681 [-0.0696, -0.0666]	S/U/U
19	0.35	+0.2620 [+0.2594, +0.2645]	-0.0217 [-0.0240, -0.0192]	+0.0188 [+0.0164, +0.0213]	S/U/S
19	0.50	+0.2668 [+0.2587, +0.2737]	+0.0016 [-0.0012, +0.0043]	-0.0029 [-0.0053, -0.0006]	S/M/U
19	0.75	+0.1807 [+0.1648, +0.1999]	-0.0111 [-0.0165, -0.0049]	-0.0619 [-0.0670, -0.0560]	S/U/U

Table 2: Condition-count study at $n = 19, s = 0.50$. Mean and worst are held-out profile TVD. The shared model keeps 190 phases and 2,000 updates.

K	regime	train/condition	validation/condition	mean	worst	phases	shared updates	reference updates
2	fixed per condition	10,000	2,000	0.3819	0.5170	190	2,000	4,000
3	fixed per condition	10,000	2,000	0.3402	0.4489	190	2,000	6,000
5	fixed per condition	10,000	2,000	0.1673	0.2917	190	2,000	10,000
9	fixed per condition	10,000	2,000	0.1595	0.2947	190	2,000	18,000
2	fixed total	9,000	1,800	0.3819	0.5166	190	2,000	4,000
3	fixed total	6,000	1,200	0.3399	0.4488	190	2,000	6,000
5	fixed total	3,600	720	0.1645	0.2932	190	2,000	10,000
9	fixed total	2,000	400	0.1622	0.2936	190	2,000	18,000

Independent model and target draws give the standard unbiased U-statistic. All random streams for masks, model samples, target batches, and validation are derived separately and recorded in the campaign artifacts.

B Gaussian protocol and complete factorial summary

For each n, width factor s, and condition c, the target shell law is a truncated discrete Gaussian. Its untruncated location is calibrated so that the truncated mean is exactly $n(0.15 + 0.70c)$, and its nominal width is $s\sqrt{n}$. Conditional and no-condition models share the complete two-local generator bank, initialization, datasets, optimizer seed, minibatches, and estimator streams. Single-condition references use condition-tagged streams.

The factorial contains $4 \times 3 \times 10$ seed cells and 840 atomic fits. The condition-count study adds 180 fits for a total of 1,020. Training uses 10,000 strings and validation uses 2,000 strings per condition in the fixed-per-condition regime. The fixed-total regime divides 18,000 training and 3,600 validation strings across conditions using prefixes of the same master pools. Held-out conditions have no sample role in the stored datasets.

Table 1 reports every size–width profile-TVD effect. Positive no-condition effects favor conditioning. Positive adjacent-reference or mixture effects would favor the shared model.

Table 2 shows the condition-density result used in the main text. The full per-condition rows, worst-condition values, paired intervals, and exact probability vectors remain in the machine-readable report.

Profile TVD intentionally quotients out within-shell variation. The exact raw-bitstring report shows that low profile error does not imply a uniform law inside each shell. We retain the raw probabilities and all secondary metrics in the release artifacts rather than reproducing hundreds of diagnostic rows in the PDF.

Table 3: All paired Curie–Weiss v3 effects.

comparison	effect [95% interval]	status
cold_no_condition_bitstring_tvd	+0.209 [+0.202, +0.215]	supported
cold_no_condition_profile_tvd	+0.212 [+0.205, +0.219]	supported
conditional_r5_bitstring_tvd_held_out	+0.128 [+0.123, +0.134]	supported
conditional_r5_bitstring_tvd_training	+0.140 [+0.135, +0.145]	supported
conditional_r5_profile_tvd_held_out	+0.121 [+0.116, +0.127]	supported
conditional_r5_profile_tvd_training	+0.134 [+0.129, +0.140]	supported
critical_no_condition_bitstring_tvd	+0.056 [+0.048, +0.065]	supported
critical_no_condition_profile_tvd	+0.038 [+0.030, +0.047]	supported
r5_vs_mixture_bitstring_tvd_held_out	+0.008 [+0.002, +0.013]	supported
r5_vs_mixture_profile_tvd_held_out	-0.011 [-0.017, -0.005]	unsupported
r5_vs_profile_oracle_held_out	+0.015 [+0.011, +0.019]	supported
r5_vs_raw_oracle_held_out	+0.036 [+0.031, +0.041]	supported
r5_vs_reference_bitstring_tvd_training	+0.008 [+0.003, +0.013]	supported
r5_vs_reference_profile_tvd_training	-0.020 [-0.024, -0.015]	unsupported

Table 4: Ancilla development study. Entries are mean held-out TVD.

r	model	profile TVD	raw TVD
0	conditioned	0.208	0.321
0	no condition	0.221	0.298
1	conditioned	0.166	0.213
1	no condition	0.227	0.261
3	conditioned	0.111	0.128
3	no condition	0.155	0.175

C Curie–Weiss protocol and results

For $s_i = 2x_i - 1$, $M = \sum_i s_i$, and $n = 8$, the exact target is

$$p_\beta(x) = Z_\beta^{-1} \exp\left(\frac{\beta M^2}{2n}\right). \tag{18}$$

The campaign checks normalization, permutation invariance, and spin-flip invariance before writing a dataset. Each of ten dataset archives contains 10,000 training and 2,000 validation strings at the six training temperatures. Exact laws at held-out midpoints enter only after checkpoint selection.

The v3 broadcast map is piecewise linear. Its knots were chosen from development screens before production and are recorded in the manifest. The map is not presented as target-independent. The production model uses the spin-flip two-branch state on eight visible and five ancilla inputs, applies a complete two-local 13-qubit IQP unitary, and sums the exact joint law over all ancilla outcomes. The encoding family and map were frozen before the 80 production fits.

Table 3 gives the complete aggregate comparison set. Temperatures are averaged within seed before bootstrapping. Positive effects favor the named conditioned model. The shared model matches the adjacent mixture and improves on the target-aware adjacent oracle in this campaign; both profile and raw TVD improve relative to no conditioning.

The five-ancilla choice followed a separate three-seed development study at $r \in \{0, 1, 3\}$. It motivated adding latent capacity but did not test $r = 5$, establish an optimal ancilla count, or isolate mixed-state rank from parameter count.

C.1 Historical v2 matched-capacity diagnostic

This diagnostic predates the v3 spin-flip campaign. It compares the v2 product-input ancilla model with a visible-only model containing every nonempty one-, two-, and three-qubit diagonal generator

Table 5: Matched-capacity paired effects. Positive values mean lower TVD for the five-ancilla model.

region	profile TVD	raw TVD
held out	+0.067 [+0.035, +0.095]	+0.144 [+0.105, +0.185]
critical	+0.118 [+0.082, +0.146]	+0.184 [+0.145, +0.218]
cold	+0.007 [-0.032, +0.048]	+0.082 [+0.033, +0.127]

on eight visible qubits:

$$\binom{8}{1} + \binom{8}{2} + \binom{8}{3} = 8 + 28 + 56 = 92. \tag{19}$$

This is one phase more than the five-ancilla complete two-local model. The control reuses the v2 datasets, product-input angle map, support-hashed initialization, estimator streams, 2,000-step optimizer, validation checkpoint rule, and exact evaluation grid. It does not reuse the v3 input state and is therefore neither a capacity match for v3 nor evidence that ancillas cause the v3 improvement. Critical aggregates average $\beta \in \{0.9, 1.0, 1.1\}$; cold aggregates average $\beta \in \{1.3, 1.4, 1.5\}$.

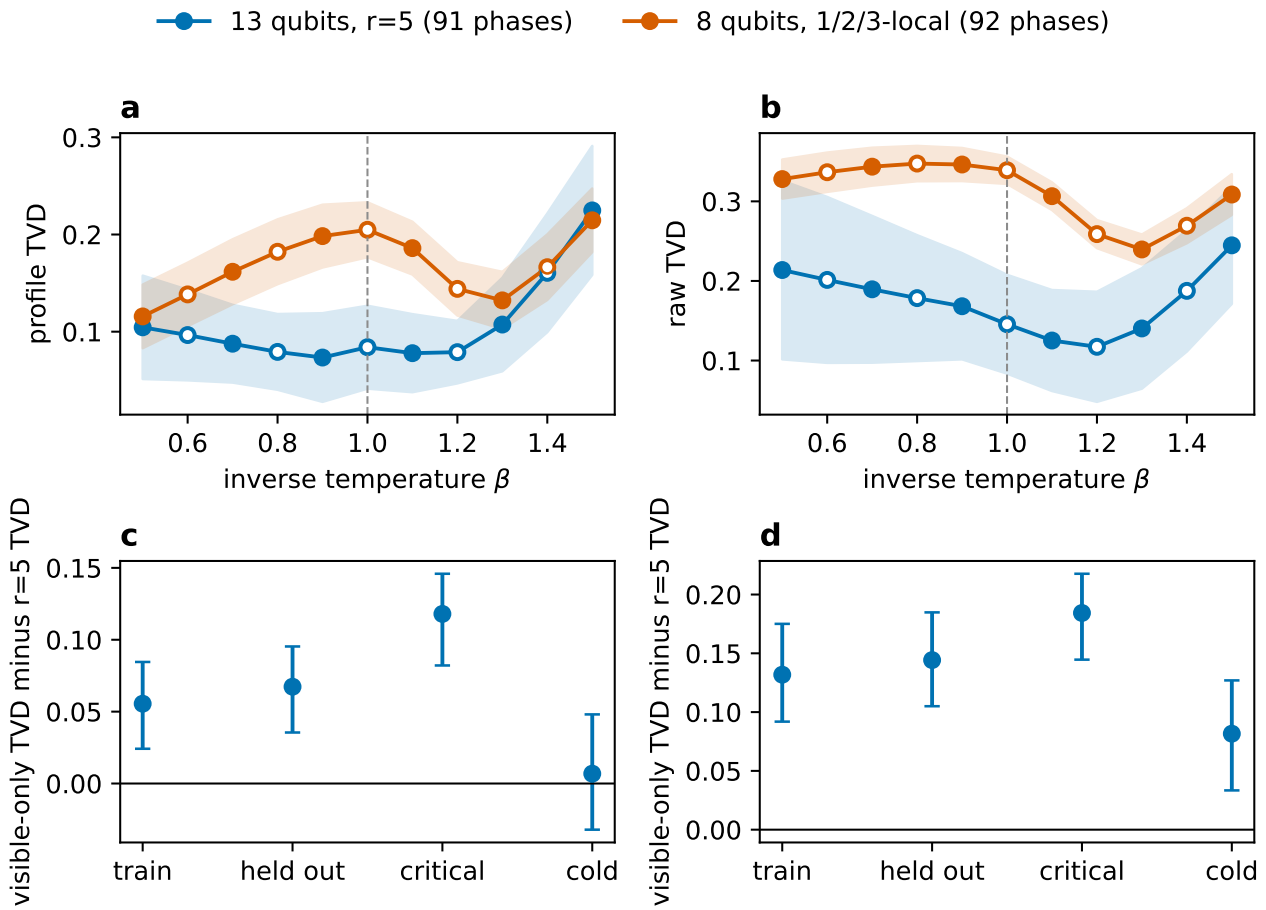


Figure 4: Historical v2 matched-capacity diagnostic. Mean profile and raw TVD and paired regional effects compare the 91-phase v2 product-input ancilla model with the 92-phase visible-only model. Positive effects favor the v2 ancilla model. This is not a matched control for the v3 spin-flip model.

D Exploratory symmetry screens

These post-core screens test whether matching an input state's symmetry to a target can improve finite-system fits. They were designed after the core campaigns and are not confirmatory evidence for the v3 result.

Two-block Z_2^2. A two-block Curie–Weiss target $p(s) \propto \exp[\beta(M_A^2+M_B^2)/16]$ on two eight-spin blocks is invariant under independently flipping each block. The tested input is a product of two two-branch states, $|\phi\rangle = (|p_A\rangle^{\otimes 8} + X^{\otimes 8}|p_A\rangle^{\otimes 8}) \otimes (|p_B\rangle^{\otimes 8} + X^{\otimes 8}|p_B\rangle^{\otimes 8})$. Across ten matched seeds it outperforms the product and single-Z_2 inputs on the reported held-out metrics. This ordering is a screen result, not a general theorem about symmetry-matched encodings.

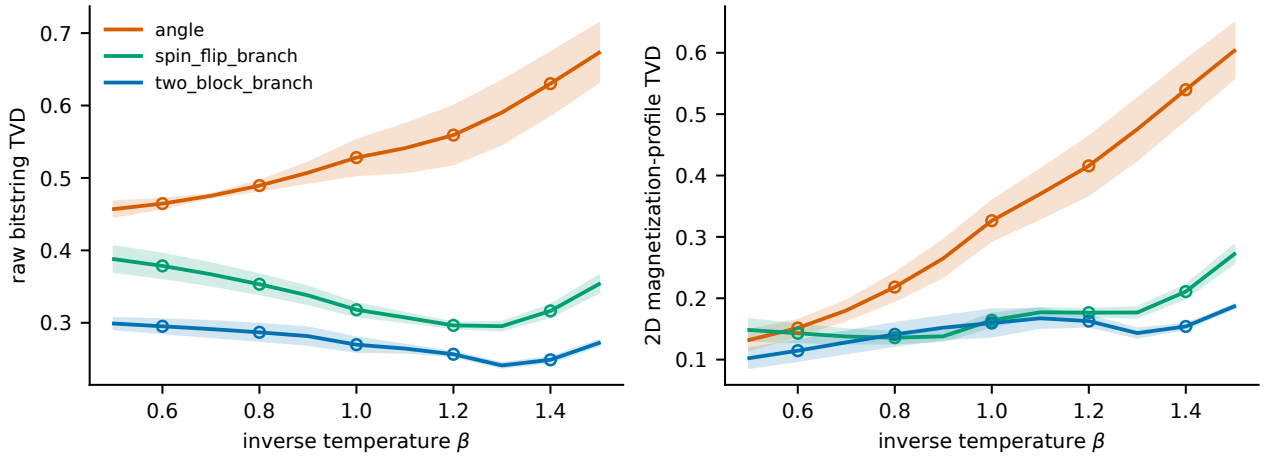


Figure 5: Exploratory two-block Z_2^2 screen. Held-out raw and two-block magnetization-profile TVD for product, single-Z_2, and two-block-Z_2^2 inputs. Bands are one standard error across ten matched seeds; hollow markers denote held-out temperatures.

Asymmetric exchangeable targets. A radial-shell state with a frozen quartic shell law accepts two coordinates (c_1, c_2) and preserves permutation symmetry without enforcing spin flip. It improves the weight-profile and permutation-averaged raw TVD in the reported screen, while its unprocessed raw TVD is worse. The product and two-branch baselines receive only c_1, whereas the radial model receives both c_1 and c_2; the result therefore does not isolate symmetry matching from access to an additional condition coordinate.

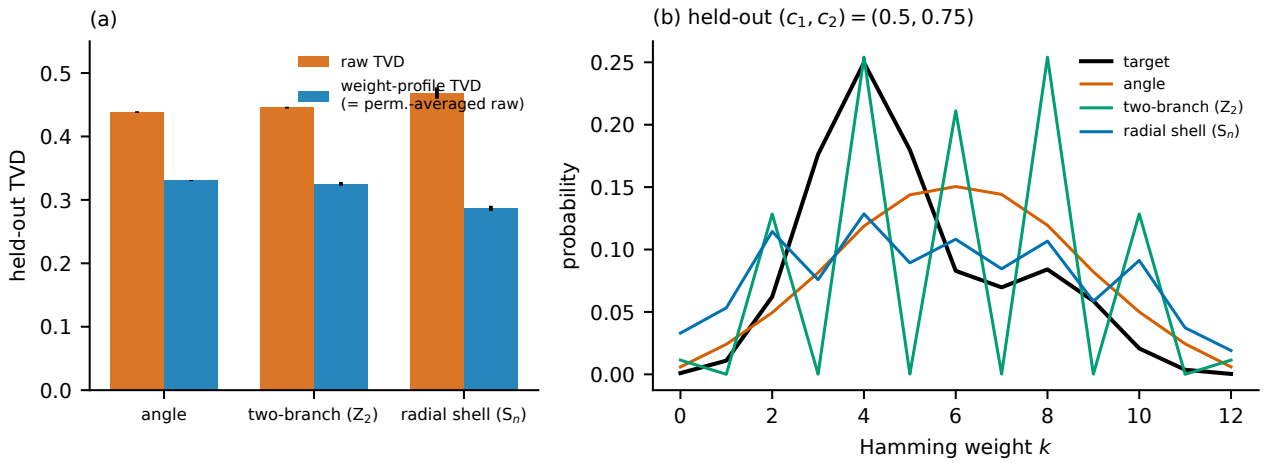


Figure 6: Exploratory asymmetric radial-shell screen. Held-out raw and weight-profile TVD and representative weight profiles are shown for the three input families. The unequal condition information makes this a feasibility screen rather than a controlled encoder comparison.

E Archived encoding comparison

We tested two extensions of the single-angle product encoding on the $n = 19, s = 0.50$ Gaussian task: noncommuting single-qubit reuploading and two-qubit block-product inputs. None improved the tuned production encoder. The study is supplementary because it followed a different development protocol and its alternatives were not used in either production campaign.

Table 6: Archived matched encoding comparison on the $n = 19, s = 0.50$ Gaussian task. Lower is better. The paired profile effect is relative to the single-angle baseline; positive values are worse.

encoding	structure	held-out profile TVD	held-out raw TVD	paired profile effect
single angle	$L = 1$	0.062 ± 0.004	0.348 ± 0.006	baseline
noncommuting	$L = 2$	0.072 ± 0.009	0.374 ± 0.027	$+0.011$ $[+0.004, +0.017]$
block ZZ	$\kappa = \pi/8$	0.070 ± 0.008	0.371 ± 0.020	$+0.009$ $[+0.003, +0.015]$
block ZZ	$\kappa = \pi/4$	0.071 ± 0.014	0.398 ± 0.009	$+0.009$ $[-0.001, +0.017]$

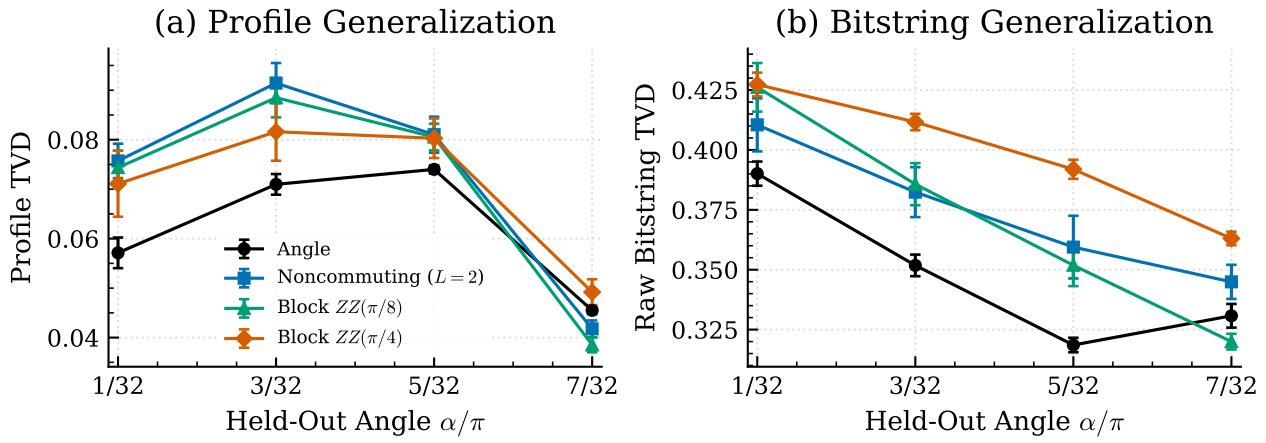


Figure 7: Archived input-encoding comparison. The left panel reports held-out conditional MMD and the right panel reports raw profile TVD for the four tested architectures. Lower is better.

F Finite estimator audit

The separate estimator audit contains 21 deterministic synthetic cases over qubit count, generator count, condition count, and condition dimension. It uses bounded-degree and complete two-local generator families at $n = 4$ and $n = 8$. Exact references are used where feasible; larger cases compare repeated estimator draws. The audit records expectation and gradient-estimator errors, runtime, memory, seeds, and content digests.

This is finite consistency and resource evidence only. It does not support an asymptotic scaling claim, a statement about raw gradient magnitude, convergence of the optimizer, or absence of barren plateaus. No full training run is charged to this audit.

G Negative and boundary studies

An eight-qubit pilot attempted joint conditioning on Gaussian location and width. Its frozen design required adequate single-condition representability before interpreting shared-model performance. Two narrow-width cells failed that screen, so the study stopped. It supplies no evidence for two-coordinate conditional generation and is retained as a negative result.

Older topology and ancilla studies remain archived under their original protocols. They are not pooled with either production campaign. The two symmetry screens above are supplementary exploratory evidence. Permutation twirling is included only as target-symmetry-aware post-processing; it is not a symmetry learned by the circuit.

H Campaign integrity and reproduction

The Gaussian and Curie–Weiss stores bind every run identity to a manifest, role-tagged dataset, estimator streams, source digest, environment, code revision, checkpoint, exact probability archive, invocation, runtime, and peak memory. Stores reject undeclared identities, duplicate completion, and silent reuse of partial results. Report, claim, table, and figure generators require complete campaign state.

The Gaussian dependency closure includes a content-addressed snapshot of the historical execution source. In particular, the archived `encoding.py` has SHA-256 `0ec0d9d2903fd44684841721c9ffac9543f2c7f883e28f8ada09bb318e258338`. The current live encoder is not substituted for that historical source.

The principal local reproduction commands are:

```
uv sync --frozen
uv run pytest -q
uv run ruff check src scripts tests
uv run python scripts/audit_gaussian_release.py
uv run python scripts/audit_curie_weiss_ancilla_release.py
uv run python scripts/run_curie_weiss_capacity_control.py run
uv run python scripts/run_curie_weiss_capacity_control.py summarize
(cd paper && latexmk -pdf main.tex)
```

Exhaustive per-seed and per-condition tables remain in JSON, NPZ, and generated TeX artifacts in the local review package. Keeping them machine-readable preserves the audit trail without turning the supplement into a second report.